



NVIDIA L40S

Unparalleled AI and graphics performance for the data center.

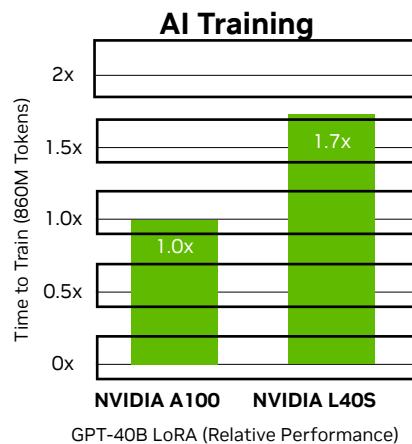


Generative AI is fueling transformative change, unlocking a new frontier of opportunities for enterprises across every industry. To transform with AI, enterprises need more compute resources, greater scale, and a broad set of capabilities to meet the demands of an ever-increasing set of diverse and complex workloads.

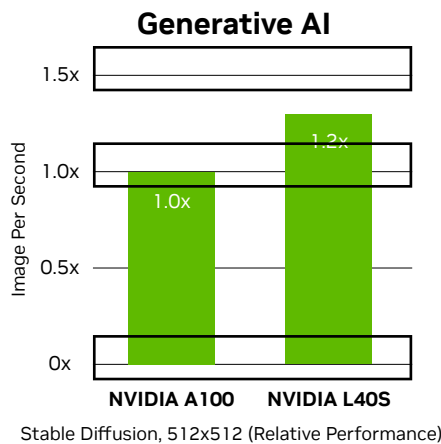
The NVIDIA L40S GPU is the most powerful universal GPU for the data center, delivering end-to-end acceleration for the next generation of AI-enabled applications generative AI and model training and inference to 3D graphics, rendering, and video applications.

Accelerate Next-Generation Workloads

- > Generative AI
- > Large language model (LLM) training and inference
- > NVIDIA Omniverse™ Enterprise
- > Rendering and 3D graphics
- > Streaming and video content



Fine-tuning LoRA (GPT-40B): global train batch size: 128 (sequences), seq-length: 256 (tokens). NVIDIA HGX™ A100 8 GPU vs. Two systems with 4x L40S GPUs. Performance based on prerelease build, subject to change.



Stable Diffusion v2.1. Relative speedup for 512 x 512 resolution image generation. NVIDIA HGX A100 8 GPU vs. Two systems with 4x L40S GPUs. Performance based on prerelease build, subject to change.

Powered by the NVIDIA Ada Lovelace Architecture

Fourth-Generation Tensor Cores

Hardware support for structural sparsity and optimized TF32 format provides out-of-the-box performance gains for faster AI and data science model training. Accelerate AI-enhanced graphics capabilities with DLSS to upscale resolution with better performance in select applications.



Thermal	Passive
Virtual GPU (vGPU) Software Support	Yes
vGPU Profiles Supported	See the virtual GPU licensing guide
NVENC NVDEC	3x 3x (includes AV1 encode and decode)
Secure Boot With Root of Trust	Yes
NEBS Ready	Level 3
MIG Support	No
NVIDIA® NVLink® Support	No

* With sparsity

Ready to get started?

To learn more about the NVIDIA L40S, visit
www.pny.com/l40s

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